

Project Charter

Project Name	Supplier Disruption Radar Agent
Course	APS1013
Professor	Stephen Armstrong
Team Name	Chain Signals
Team Members	Ahmad Khan, Armin Sariri, Benjamin Luo, Dasha Moskvitina, Manni Zhang, Selim Akef, Tobias Udorji, Xiaoming Chen

1 CHARTER INTRODUCTION

1.1 REVISION HISTORY

Version	Date	Author	Description of Changes
1.0	2026/05/25	Chain Signals	Baseline version

1.2 EXECUTIVE SUMMARY

Modern supply chains are increasingly vulnerable to disruptions caused by geopolitical instability, natural disasters, transportation bottlenecks, supplier instability, cyberattacks, and rapidly shifting market demand. Organizations often struggle to identify these disruptions early enough to prevent operational delays, inventory shortages, and increased costs. To address this challenge, our team is developing a Supplier Disruption Radar Agent, a multi-agent AI framework designed to automate the transition from raw public disruption signals into structured supply chain risk analysis and mitigation planning. By leveraging large language models and a modular Python-based architecture, the system aims to provide organizations with visibility into emerging supply chain threats and enable operational decision-making.

This report details the project description, goals and scope for the Supplier Disruption Radar Agent. It outlines the intended functionality of the system, including the generation of disruption cards, mitigation playbooks, and validation plans derived from public supply chain disruption signals. The report also defines the current constraints and flexibility considerations that will guide ongoing stakeholder discussions regarding system inputs, outputs, and overall project requirements.

In addition, this report presents the key project deliverables, division of labour among team members, development timeline, and final presentation structure that will guide the implementation and demonstration of the proposed solution.

2 PROJECT DESCRIPTION

Our project, the Supplier Disruption Radar Agent, is an AI-powered supply chain intelligence platform designed to help organizations proactively identify and assess operational risks impacting specific industry supply bases. For the current prototype, the system focuses on live news retrieval and AI-driven disruption analysis. Using industry-specific search profiles, the agent monitors public news feeds for potential disruption signals related to events such as geopolitical instability, labor strikes, natural disasters, logistic bottlenecks, supplier shutdowns, export restrictions, and material shortages. Once a relevant signal is retrieved, the platform uses large language models to evaluate the event in the context of the selected supply base, classify the disruption using a structured taxonomy, calculate a risk score, and generate standardized disruption cards containing affected regions, suppliers, impacted components, estimated time-to-hit, and operational summaries. The project demonstrates how AI agents can transform unstructured public information into supply chain intelligence that supports risk monitoring, prioritization, and mitigation planning.

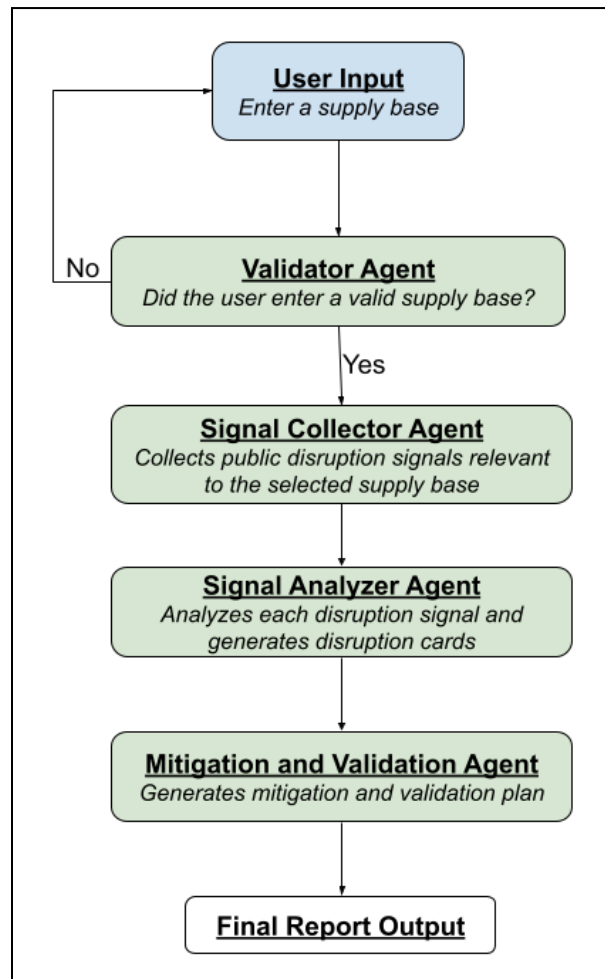


Figure 1: Baseline System Architecture

Seen in Figure 1, the system architecture is composed of four main AI-driven components. First, the Supply Base Validation Agent checks whether the user's input represents a real industry, supplier ecosystem, or procurement category. Next, the Live Signal Retrieval Agent searches public news feeds using industry-specific keywords to identify headlines that may indicate supply chain disruption. The Disruption Analysis Agent then evaluates each signal in the context of the selected supply base, classifies it using a disruption taxonomy, estimates likelihood, impact, time-to-hit, and assigns a risk score. Finally, the Mitigation Planning And Validation Agent translates each disruption card into recommended actions such as alternate sourcing, inventory buffers, logistics rerouting, supplier communication, and escalation steps. Together, these agents convert raw public news into structured disruption cards and practical response guidance.

At the current stage of development, the team is prioritizing modularity and flexibility in system design. The specific data ingestion methods, supported file formats, reporting structures, and integration requirements are expected to be refined through ongoing discussions with project stakeholders.

3 PROJECT GOALS

The primary goal of this project is to demonstrate the feasibility of an AI-driven Supplier Disruption Radar Agent that can transform public disruption signals into actionable supply chain risk intelligence. The system aims to help organizations move from reactive crisis management toward proactive, data-driven disruption monitoring, risk assessment, and mitigation planning.

To achieve this goal, the project will pursue the following objectives:

1. Define a target industry supply base to provide a focused environment for the agent's risk sensing and monitoring activities.
2. Establish a comprehensive signal taxonomy to categorize external disruptions identified through public data sources, including geopolitical instability, labour strikes, natural disasters, logistics bottlenecks, supplier shutdowns, export restrictions, and material shortages.
3. Develop a structured scoring model to evaluate disruption signals based on likelihood, impact, and time-to-hit.
4. Build a functional multi-agent demo capable of validating supply chain inputs, retrieving public disruption signals, analyzing operational risk, and generating response recommendations.
5. Automate the creation of standardized disruption cards containing disruption classifications, risk scores, impacted regions, suppliers, components, estimated time-to-hit, and operational impact summaries.
6. Generate industry-specific mitigation playbooks tailored to the operational dependencies and vulnerabilities of the selected supply base, including recommendations for alternate sourcing, inventory buffers, logistics rerouting, supplier communication, and escalation steps.
7. Incorporate both AI-driven analysis and human validation processes to assess disruption credibility, improve output reliability, and monitor evolving risk developments.

8. Conduct an accuracy assessment using real-world test cases to evaluate the reliability of the agent's findings, risk scores, and recommendations.
9. Build the platform with flexible input and output structures that can evolve based on stakeholder feedback, business requirements, and future integrations.

4 SCOPE

4.1 IN SCOPE

The project scope includes the development of a Python-based application that processes supply chain inputs and generates structured disruption analysis using a series of OpenAI-powered agents.

The implementation will support the following core functions:

- Validate that user-provided supply chain inputs conform to the input format, structure, and requirements agreed upon with stakeholders during the design process before processing begins
- Collect and analyze public disruption signals relevant to a specified supply base
- Generate structured disruption cards containing risk assessments and operational impact summaries
- Generate mitigation playbooks focused on alternate sourcing, inventory management, logistics planning, and communication strategies
- Generate validation plans describing how disruption credibility and ongoing risk exposure may be assessed

The team intends to maintain flexibility regarding the system's final input and output formats until further stakeholder discussions occur. Potential inputs may include direct user prompts, structured CSV datasets, spreadsheets, or externally supplied reports. Potential outputs may include terminal reports, structured JSON, text-based summaries, PDFs, or other Python-supported export formats.

The project also includes the investigation of practical methods for leveraging large language models within supply chain risk analysis workflows. The accuracy and quality of generated disruption signals, mitigation recommendations, and validation plans will depend on the capabilities and limitations of the underlying AI models and publicly available data sources.

As such, the system should be viewed as a decision-support and risk-awareness tool rather than a guaranteed source of factual or operational truth.

4.2 OUT OF SCOPE

The project scope does not include integration into Deloitte's internal production systems or infrastructure. The current agreement is that the application will execute locally on developer-managed machines for demonstration and evaluation purposes.

The team has elected to use OpenAI’s GPT5.5 model as the primary AI platform for the project [1]. While the team remains open to evaluating alternative OpenAI model variants where appropriate, comparative benchmarking across multiple external AI vendors (e.g., Anthropic Claude, Google Gemini, etc.) is considered outside the current project scope. Expanding the project to include cross-vendor AI evaluation would introduce substantial technical overhead, increased validation complexity, and a significant increase in API-related costs beyond the currently proposed project budget of 50\$. Additionally, rigorous comparative accuracy testing across multiple model providers would require extensive manual validation effort that exceeds the available project timeline and resources.

The team also does not agree to use private or paywalled data sources during the collection of public disruption signals. The project is specifically intended to evaluate the feasibility of leveraging publicly accessible information for supply chain disruption monitoring. Incorporating proprietary datasets, subscription-based intelligence platforms, or confidential enterprise data would introduce additional licensing and access management that fall outside the scope of the current project. Additionally, restricting the system to publicly available data improves reproducibility, accessibility, and transparency during system validation and stakeholder evaluation.

5 KEY DELIVERABLES

Project Deliverable 1: Project Charter	
Description:	A foundational document outlining the team's project approach, roles and responsibilities, and proposed timelines presented in a Gantt chart format. It clearly defines the project scope (in-scope and out-of-scope) and details the initial framework and structure for the final report and Generative AI agent demo.
Due Date:	May 25, 2026
Project Deliverable 2: Preliminary Project Presentation	
Description:	A midway progress update and draft review shared with Deloitte mentors. This presentation serves as a quality checkpoint to align the team’s trajectory with Deloitte’s professional standards, showcasing early development of the signal taxonomy, scoring models, and preliminary agent architecture before finalization.
Due Date:	June 5, 2026
Project Deliverable 3: Project Presentation	
Description:	A final executive presentation delivered to the Deloitte mentorship team (scheduled for 3:00 PM–5:00 PM ET). It comprehensively communicates the project’s methodology, the technical build of the AI agent, strategic insights, and the final business value of the supply chain disruption solution.
Due Date:	June 11, 2026

Project Deliverable 4: Project Report	
Description:	The final, comprehensive technical and business report completing the project. It encapsulates the functional Agent Demo, a Signal Taxonomy + Scoring Model (evaluating likelihood, impact, and time-to-hit), a portfolio of 10–15 standardized disruption cards, and a robust Mitigation Playbook detailing alternative logistics, inventory strategies, and validation plans.
Due Date:	June 19, 2026

6 DIVISION OF LABOUR

Project Workstreams	Ahmad	Armin	Ben	Dasha	Manni	Selim	Tobias	Xiaoming
Project Management & Coordination								
1. Project Charter								
2. Project Coordination								
Market & Academic Research Track								
3. Supply Chain Practices								
4. Industry Sourcing & Datasets								
5. Threat Taxonomy & Risks								
Technical Development Track								
6. System Architecture & Pipeline								
7. Data Engineering & Validation								
8. AI Agent Core Engineering								
9. Disruption Cards & Scoring								
10. UI/UX Design								
11. System Validation/Verification								
Strategic Synthesis & Delivery								
12. Mitigation Playbook								
13. Final Presentation								
14. Project Report								

Write	Review
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7 PROJECT REFERENCES

[1] OpenAI, “API Pricing,” OpenAI, 2026. [Online]. Available: <https://openai.com/api/pricing/>.

8 GANTT CHART

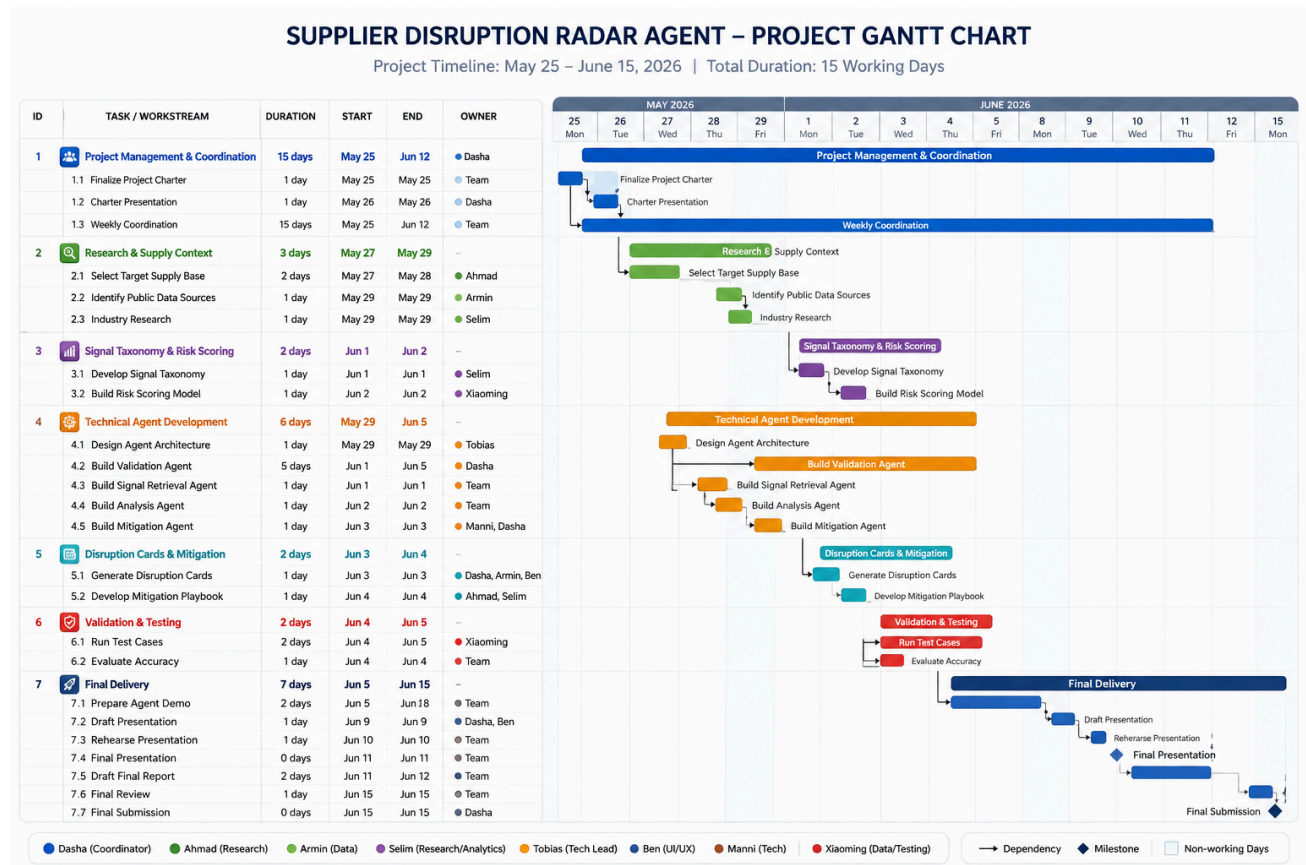


Figure 2: Proposed Gantt Chart

The Gantt chart outlines a 15-day project (May 25–June 15, 2026), showing phases, dependencies, and team responsibilities.

The project starts with Project Management and Coordination, ensuring the charter is completed, presented, and activities are coordinated throughout.

The Research and Supply Context phase defines the supply base, data sources, and industry insights, forming the project foundation.

Next, Signal Taxonomy and Risk Scoring develops disruption signals and a risk model from research outputs.

The core Technical Agent Development phase designs the system architecture and builds validation, retrieval, analysis, and mitigation agents in parallel.

The Disruption Cards and Mitigation phase converts outputs into actionable insights and a mitigation playbook.

This is followed by Validation and Testing, where performance and accuracy are evaluated.

Finally, the Final Delivery phase includes demo preparation, presentation (June 11), and final report submission (June 15).

Overall, the Gantt chart ensures clear sequencing, dependencies, and coordination for efficient project delivery.

9 FINAL PRESENTATION STRUCTURE

Module & Timing	Core Content	Description
Module 1: Strategic Context (~3 Mins / 15%)	The Supply Chain Challenge	Frames the chosen industry supply base and explains how unmonitored public signals create critical operational vulnerabilities.
	Project Scope	Defines the core objectives of the agent build and establishes project boundary lines to manage executive expectations.
Module 2: Methodology & Taxonomy (~6 Mins / 30%)	Signal Taxonomy	Provides a technical breakdown of the open/public data sources selected and the structural taxonomy of defined disruption signals.
	Predictive Scoring Model	Explains the underlying logic, weightings, and calculations driving the evaluation metrics: Likelihood, Impact, and Time-to-Hit.
Module 3: Technical Agent Build (~7 Mins / 35%)	Agent Architecture	Shows a high-level schematic architectural overview of how the Generative AI agent processes raw public signals into structured data.
	Live/Recorded Agent Demo	Conducts an operational walkthrough of the functional agent successfully ingesting real-time data and generating outputs.
	Disruption Cards Showcase	Reviews selected cards from the 10–15 standardized portfolio, highlighting automated risk scores, impacted parts, and mitigation triggers.
Q&A Session	Technical Appendix	Backs up the presentation with additional slides mapping database lineages, code snippets, and granular source data.

